



DEALER BEST PRACTICE SERIES

Contamination Control Using Plugs & Caps

Application Maintenance Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

Dealers can improve their Contamination Control Program by expanding the use of plastic caps and plugs. During repair, every machine component or system that is opened/disassembled has to be protected to avoid contamination. Caterpillar supplies a variety of plastic caps and plugs through the normal parts system. This practice leads to Contamination Control practices that have proven to effectively contribute to component durability.

Our task as maintenance personnel is to embrace and implement the correct repair/service procedures that will ensure the best possible results in terms of repair quality and efficiency.

A dealer and customer partnered to launch a strong project to introduce and implement all necessary Contamination Control initiatives. This effort started to give positive results that can be seen in both component durability and failures reduction. Implementing this aspect of Contamination Control has aided this dealer in achieving 5 Stars in the Caterpillar Contamination Control Audit program during the last two years, making it the first mining maintenance operation to achieve this recognition.

2.0 Best Practice Description

This Best Practice facilitates the supply of plastic plugs and caps (P&C) by locating them close to all repairs in progress. A Latin American dealer designed and fabricated two portable cabinets that can be easily moved to the repair bays. The cabinets are stocked with a complete set of plastic plugs and caps for quick access.



General view of one of the P&C cabinets

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Plastic plugs and caps are stored in plastic bags and the cabinet doors are sealed to eliminate any external contamination.



All P&C compartments are identified and replenishment is the responsibility of the on-site warehouse.

When a repair is in progress, the technicians move the cabinet close to the work area to have the P&C close and easily available when needed. Whenever a compartment/system/component is open, the appropriate plastic plugs and caps are selected and installed over the openings.

Warehouse personnel check the P&C Cabinet quantities as part of their daily check routine.

All shop personnel follow this practice and its use is constantly supervised through the Supervisors, Staff Personnel, and Project Manager.

3.0 Implementation Steps

1. Based on the magnitude of your operation (number of bays and typical workload) define the number of cabinets needed.

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2. Fabricate the plugs & caps cabinet. The cost to fabricate each cabinet locally was approximately US \$1,500
3. Define the set of plastic plugs and caps to maintain in the cabinet, both part numbers and quantity. You can follow the recommendations included in Section 6.0.
4. Order a complete set of Plugs & Caps. Define the inventory limits for replenishment. Make the warehouse personnel responsible for replenishment.
5. Always use new Plugs & Caps. Experience shows that recycling caps is not worth the effort and introduces possibilities for contamination.
6. Prepare the cabinets and stock with the plugs and caps. Assign them to the shop.
7. Train personnel in the use of the plugs and caps and the cabinet system.
8. Instruct supervisors to reinforce the need to properly utilize these protective devices.
9. Make shop supervisors responsible for the maintenance of the carts and their correct use.
10. Include the carts in all Shop Facilities & Equipment Audits.

4.0 Benefits

- Contamination Control
 - o Ensures proper storage of open machine systems / components.
 - o Eliminate oil spillages, improve cleanliness of working area.
 - o Contributes to component durability.
- Work quality - by having plugs and caps easily visible and available, reduces occurrences of work being performed without caps.
- Work Efficiency - contributes to improve work efficiency (Mean Time to Repair). Items are available for immediate use and don't have to be searched for.
- Environmental Control - prevents oil spills, contributes to a better and cleaner environment.
- Image - contributes to enhance the dealer image, shows professional work done following all recommended procedures.
- Easy to manage. This supply system for the plugs and caps is very visible and easy to supervise. Everyone can see it in the shop.

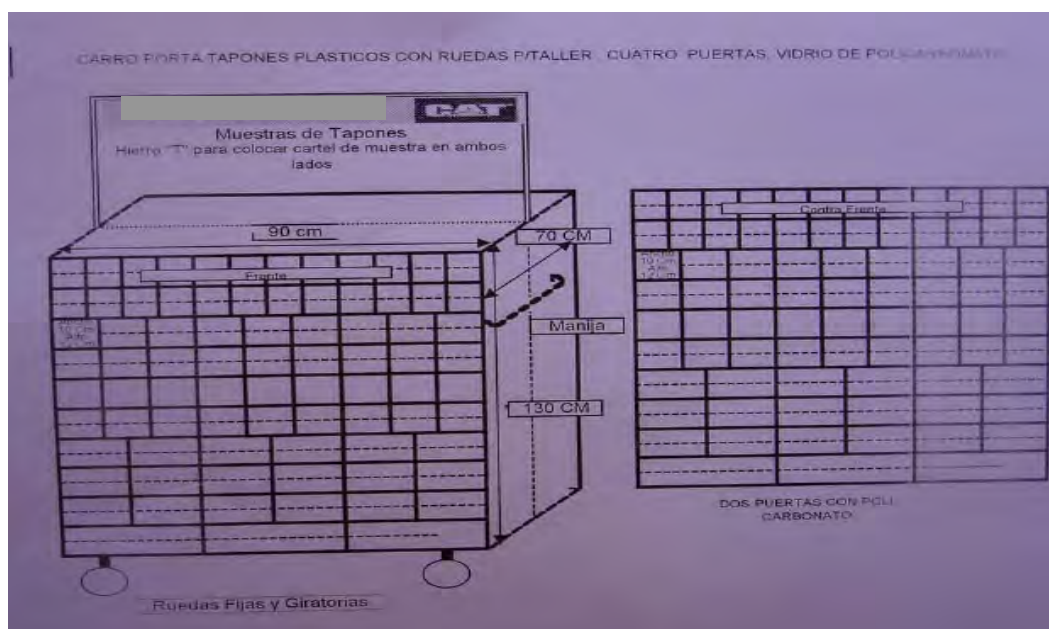
5.0 Resources Required

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- Fabricated portable cabinet. The cost of each cart fabricated locally in Argentina was US \$1,500. Use this cost as reference only. The number of carts to fabricate depends on the equipment fleet size and the frequency of the repairs/services that are performed in the shop.
- One complete set of plastic plugs & caps. The whole recommended set has a list cost of US \$1,443. See Section 6.0 for the complete parts list.
- Local warehouse staff or parts specialist for replenishment and inventory control of the cabinets.

6.0 Supporting Attachments / References

1. Drawing used for original fabrication:



2. Recommended parts list for stocking the plugs & caps cabinet:

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Columns headings are: Unit Price, Total Price, Part Number, Quantity, Thread Size, and specific dimensions ("Diam. Tubo" refers to Tube Diameter)

PRECIO UNITARIO	TOTAL	PART N°	QTY	TAMANO HILO	A (mm)	B MM	REF.
\$0.02	\$5.00	9U7120	250	7/16 - 20	12	11	4
\$0.02	\$5.00	9U7121	250	1/2 - 20	12	12	5
\$0.02	\$5.00	9U7122	250	9/16 - 18	12	14	6
\$0.02	\$5.00	9U7123	250	11/16 - 16	12	17	7
\$0.04	\$10.00	9U7124	250	3/4 - 16	14	19	8
\$0.03	\$7.50	9U7125	250	13/16 - 16	14	20	9
\$0.04	\$10.00	9U7126	250	7/8 - 14	17	22	10
\$0.03	\$7.50	9U7127	250	1" - 14	19	25	11
\$0.04	\$10.00	9U7128	250	11/16 - 12	19	27	12
\$0.05	\$12.50	9U7129	250	13/16 - 12	19	30	13
\$0.04	\$4.00	9U7130	100	16/16 - 12	19	33	16
\$0.05	\$5.00	9U7131	100	17/16 - 12	19	36	17
\$0.05	\$5.00	9U7132	100	15/8 - 12	21	41	20
\$0.08	\$8.00	9U7133	100	111/16 - 12	21	42	21
\$0.11	\$11.00	9U7134	100	17/8 - 12	24	48	24
\$0.14	\$7.00	9U7135	50	21/2 - 12	30	63	32

\$117.50

PRECIO UNITARIO	TOTAL	PART N°	QTY	TAMANO HILO	A (mm)	B (mm)	C (mm)
\$0.06	\$6.00	9U7195	100	1/8 - 28	11	16	4
\$0.06	\$6.00	9U7196	100	1/4 - 19	13	19	7
\$0.06	\$6.00	9U7197	100	3/8 - 19	12	23	6
\$0.13	\$13.00	9U7198	100	1/2 - 14	17	28	10
\$0.11	\$5.50	9U7199	50	5/8 - 14	18	31	13
\$0.16	\$8.00	9U7200	50	3/4 - 14	17	32	11
\$0.16	\$8.00	9U7201	50	7/8 - 14	18	38	14
\$0.27	\$13.50	9U7202	50	1 - 11	17	40	11

\$66.00

PRECIO UNITARIO	TOTAL	PART N°	QTY	TAMANO HILO	DIAM.TUBO (mm)	REF.
\$0.04	\$10.00	9U7114	250	7/16 - 20	6.4	3
\$0.05	\$12.50	9U7115	250	9/16 - 18	9.5	5
\$0.05	\$12.50	9U7116	250	3/4 - 16	12.7	7
\$0.06	\$6.00	9U7117	100	7/8 - 14	15.9	8
\$0.07	\$7.00	9U7118	100	11/16 - 12	19.1	9
\$0.12	\$12.00	9U7119	100	15/16 - 12	25.4	12

\$60.00

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PRECIO UNITARIO	TOTAL	PART N°	QTY	NUMERO DE REF.	A (mm)	B (mm)	C (mm)	D (mm)	E (mm)	F (mm)
\$0.01	\$2.50	9U7067	250	8	10	8	14	10	7	8
\$0.01	\$2.50	9U7068	250		11	9	15	10	8	9
\$0.01	\$2.50	9U7069	250	11	12	11	15	10	9	11
\$0.01	\$2.50	9U7070	250	13	14	12	18	13	10	12
\$0.02	\$5.00	9U7071	250	16	16	13	19	13	12	14
\$0.02	\$5.00	9U7072	250	17	16	14	20	13	12	14
\$0.03	\$7.50	9U7073	250		18	16	22	13	15	17
\$0.02	\$5.00	9U7074	250	22	19	17	22	16	16	18
\$0.03	\$7.50	9U7075	250	23	20	18	24	16	16	18
\$0.02	\$5.00	9U7076	250	27	22	19	26	16	17	20
\$0.04	\$10.00	9U7077	250	28	23	20	28	19	19	21
\$0.05	\$12.50	9U7078	250	31	25	22	28	16	21	23
\$0.04	\$10.00	9U7079	250	35	26	23	29	19	21	25
\$0.03	\$7.50	9U7080	250	40	28	25	32	19	23	26
\$0.03	\$7.50	9U7081	250	41	29	25	33	19	24	27
\$0.03	\$7.50	9U7082	250	42	31	28	39	19	26	30
\$0.03	\$7.50	7E8716	250	44	33	30	37	19	28	31
\$0.04	\$10.00	9U7084	250		34	31	38	19	30	33
\$0.05	\$12.50	9U7085	250	53	41	37	44	19	35	39
\$0.06	\$6.00	9U7086	100	54	41	39	44	19	37	39
\$0.05	\$5.00	9U7087	100		43	40	48	19	38	41
\$0.07	\$7.00	9U7088	100	57	48	44	52	19	43	46
\$0.06	\$6.00	9U7089	100	58	49	46	54	25	44	48
\$0.11	\$5.50	9U7090	50	68	57	57	64	13	55	56
\$0.10	\$5.00	9U7091	50	71	63	58	67	25	57	61
\$0.12	\$6.00	9U7092	50	78	71	67	76	13	66	69
\$0.19	\$9.50	6V7494	50		76	70	79	25	68	75
\$0.20	\$10.00	9U7094	50	87	88	84	92	13	82	86
\$0.24	\$12.00	9U7095	50	88	91	82	100	25	80	89
\$0.29	\$14.50	9U7096	50	94	113	107	127	25	105	111
\$0.24	\$12.00	6V7496	50	95	117	111	127	25	109	114

\$228.50

PRECIO UNITARIO	TOTAL	PART N°	QTY	TAMANO HILO	A (mm)	B (mm)	C (mm)
\$0.07	\$7.00	9U7145	100	M10 X 1,0	10	14	8
\$0.08	\$8.00	9U7146	100	M12 X 1,0	12	16	10
\$0.12	\$12.00	9U7147	100	M14 X 1,5	12	18	10
\$0.08	\$8.00	9U7148	100	M16 X 1,5	12	20	10
\$0.13	\$13.00	9U7149	100	M18 X 1,5	14	22	12
\$0.11	\$11.00	9U7150	100	M20 X 1,5	14	24	10
\$0.14	\$7.00	9U7151	50	M22 X 1,5	14	26	12
\$0.15	\$7.50	9U7152	50	M24 X 1,5	14	29	12
\$0.22	\$11.00	9U7153	50	M26 X 1,5	14	30	12

\$84.50

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PRECIO UNITARIO	TOTAL	PART N°	QTY	DIÁMETRO	A (mm)	B (mm)
\$0.02	\$5.00	9U7166	250	1/8	3	16
\$0.02	\$5.00	9U7167	250	1/4	6	22
\$0.02	\$5.00	9U7168	250	5/16	7	22
\$0.02	\$5.00	9U7169	250	3/8	10	22
\$0.02	\$5.00	9U7170	250	7/16	11	22
\$0.02	\$5.00	9U7171	250	1/2	13	22
\$0.02	\$5.00	9U7172	250	9/16	14	22
\$0.02	\$5.00	9U7173	250	5/8	16	22
\$0.04	\$4.00	9U7174	100	11/16	17	22
\$0.03	\$3.00	9U7175	100	3/4	19	22
\$0.03	\$3.00	9U7176	100	13/16	21	22
\$0.05	\$5.00	9U7177	100	7/8	22	22
\$0.04	\$4.00	9U7178	100	1"	25	19
\$0.05	\$5.00	9U7179	100	1.1/16	27	22
\$0.04	\$4.00	9U7180	100	1.3/16	30	25
\$0.07	\$7.00	9U7181	100	1.1/4	32	22
\$0.07	\$7.00	9U7182	100	1.5/16	33	22
\$0.08	\$8.00	9U7183	100	1.3/8	35	22
\$0.07	\$7.00	9U7184	100	1.7/16	37	22
\$0.08	\$8.00	9U7185	100	1.1/2	38	22
\$0.08	\$8.00	9U7186	100	1.5/8	41	25
\$0.10	\$10.00	9U7187	100	1.11/16	43	22
\$0.11	\$11.00	9U7188	100	1.3/4	44	22
\$0.12	\$12.00	9U7189	100	1.7/8	48	22
\$0.14	\$14.00	9U7190	100	2"	51	22
\$0.13	\$13.00	9U7191	100	2.1/8	54	22
\$0.17	\$17.00	9U7192	100	2.3/8	60	22
\$0.20	\$20.00	9U7193	100	2.1/2	64	22
\$0.22	\$11.00	9U7194	50	3.1/8	79	35

\$221.00

PRECIO UNITARIO	TOTAL	PART N°	CANTIDAD SUGERIDA
\$10.18	\$20.36	215-4566	2
\$10.54	\$21.08	215-4567	2
\$10.94	\$21.88	215-4568	2
\$10.94	\$21.88	215-4569	2
\$14.25	\$28.50	215-4570	2
\$13.89	\$27.78	215-4571	2
\$16.35	\$32.70	215-4572	2
\$17.48	\$52.44	215-4573	3
\$19.88	\$59.64	215-4574	3
\$16.28	\$48.84	215-4575	3
\$12.57	\$37.71	215-4576	3
\$12.70	\$38.10	215-4577	3
\$13.52	\$40.56	215-4578	3
\$17.11	\$51.33	215-4579	3
\$14.00	\$42.00	215-4580	3
\$17.83	\$71.32	215-4581	4
\$19.75	\$79.00	215-4582	4
\$19.75	\$79.00	215-4583	4
\$29.92	\$119.68	215-4584	4

\$893.80

Total Cost: US\$ 1,442.80

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7.0 Related Best Practices

None.

8.0 Acknowledgements

This Best Practice was written by:

Abelardo A. Flores
Senior Equipment Management Consultant
Caterpillar Global Mining
Flores_Abelardo@cat.com
Ph: +56-993097135

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Our special recognition to:

Victor Mato, Finning Alumbraera Project Manager.
victor.mato@finning.com.ar
Ph: +54-0835485295
Enrique Uriburu, former Finning Alumbraera Project Manager

and in the Shop & Planning area:

Victor Latof, Finning's Alumbraera Maintenance Superintendent
Jose Medina, Finning's Alumbraera Scheduler & Fluid Samples Analyst

For further details and information regarding this Best Practice, please contact the author.

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